Curriculum Vitae

Stephanie Y. Chia 賈媛

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Education

Expected Spring 2026	**Ph.D. Candidate in Ecology and Evolution**, University of Maryland (UMD)
2017	**M.S. Sustainable Business and Innovation**, Utrecht University, Netherlands
2015	**B.S. Life Science**, National Taiwan University, Taiwan Study abroad, University of British Columbia, Canada (2013)

Fellowships

2024-2026	Taiwan Ministry of Education Study Abroad Scholarship ($32,000)
2024	Smithsonian National Zoo and Conservation Biology Institute Fellowship ($3,750)
2022	NRT-INFEWS Fellowship at UMD, National Science Foundation
2021-2022	University of Maryland Dean's Fellowship ($5,000)

Research and professional experience

2021-	**Doctoral research, University of Maryland**, USA Advisor: Bill Fagan Projects: bird nest macroecology, bird morphology macroevolution, cropland optimization for water saving, spatiotemporal analysis of carnivore movement
2019-2021	**Research Assistant, Biodiversity Research Center, Academia Sinica**, Taiwan Advisor: Mao-Ning Tuanmu Projects: moth community ecology, bird altitudinal migration, bird nest traits
2017-2019	**Assistant Engineer, SGS Ltd.**, Taiwan Climate Change Innovative Actions Team Projects: climate change risk assessment, flood prediction and response, urban cooling
2015-2017	**Master's research, Utrecht University**, Netherlands Advisor: Agni Kalfagianni Thesis: Reducing food waste through waste monitoring and consumption forecast
2014	**Undergrad research, National Taiwan University**, Taiwan Project: Climate-adaptive strategies of organic tea farming Project: Real-time analysis of per-second groundwater level records

Publications

Peer-reviewed journals

Chia SY, Swain A, Josephs N, Lin L, Fagan WF. (in press). Birds that don't exist: niche pre-emption as a constraint on morphological evolution in the Passeroidea. *Ecology Letters*. (*preprint available*)

Fagan WF, Krishnan AG, Fleming CH, Sharkey E, Chia SY, Swain A, … 170+ authors. (2025). Wild canids and felids differ in their reliance on travel routes with implications for encounter dynamics. *Proceedings of the National Academy of Sciences, 122* (40) e2401042122. (cover article)

Chia SY, Wu S, Lu Y-J, Jang Y-S, Huang C-C, Shen S-F, Chen I-C, Tuanmu M-N. (2024). Mechanistic understanding of how temperature and its variability shape body size composition in moth communities. *Functional Ecology, 38*(1), 206-218. (shortlisted for Haldane Prize)

Chia SY, Fang Y-T, Su Y-T, Tsai P-Y, Hsieh C, Tsao S-H, Juang J-Y, Hung C-M, Tuanmu M-N. (2023). A global dataset of bird nest traits. *Scientific Data, 10*, 923.

Tsai P-Y, Ko C-J, Chia SY, Lu Y-J, Tuanmu M-N. (2021). New insights into the patterns and drivers of avian altitudinal migration from a growing crowdsourcing data source. *Ecography, 44*(1), 75-86.

Upcoming

Chia SY, Tsai-Chen Y, Tuanmu M-N, Fagan WF. Predation, climate, and life-history traits shape macroecological patterns of enclosed bird nests. *Preprint*. (in revision)

Chia SY, Balan R, Fagan WF. Optimizing crop distributions for water saving accounting for spatial-temporal water availability. (in prep)

Presentations

Talks

2026 Society for Integrative and Comparative Biology, Portland, OR, USA.

2025 Ecological Society of America, Baltimore, MD, USA.

2024 Mathematical Biology Seminar, Department of Mathematics, UMD.

2023 American Society of Agronomy, Crop Science Society of America and Soil Science Society of America, St. Louis, MO, USA.

2023 American Ornithological Society and Society of Canadian Ornithologists, London, ON, Canada.

2022 Belmont Forum Climate Environment and Health Early Career Researcher Workshop, virtual.

2021 Congress of Animal Behavior and Ecology, Taiwan, virtual.

Posters

2024 American Ornithological Society, Estes Park, CO, USA.

2021 Ecological Society of America, virtual.

2020 British Ecological Society, virtual.

Teaching experience

Nature at Risk: Extinction, Consequences, and Strategies (HNUH 258C), UMD
Teaching assistant (Fall 2025), Course material development (from Fall 2024 to Summer 2025)

Population Ecology (BSCI 462 / BIOL 708R), UMD
Teaching assistant (Spring 2024), Guest lectures (Spring 2025)

Human Anatomy and Physiology (BSCI 201), UMD
Teaching assistant (Spring 2022, Fall 2022, Spring 2023, Fall 2023)

Awards and grants

2024	UMD Behavior, Ecology, Evolution, and Systematics Best Graduate Research Talk
2024	Jacob K. Goldhaber Travel Grant, UMD ($400)
2024	Biological Sciences Graduate Program Travel Award, UMD ($500)
2024	Department of Biology Travel Award, UMD ($750)
2023	UMD Outstanding Graduate Teaching Assistant Award
2023	NRT-INFEWS UMD Travel Grant, NSF ($1,500)
2023	Jacob K. Goldhaber Travel Grant, UMD ($600)
2023	International Conference Student Support Award, UMD ($500)
2023	American Ornithological Society Travel Award ($560)
2021	Ecological Society of America Registration Grant

Service and outreach

2022-2025	Served on multiple graduate student organization committees, UMD BEES Program
2022	Seminar committee, NRT-INFEWS UMD Global STEWARDS Program
2019-2020	Developed educational materials and conduct public outreach activities, Academia Sinica
2018	Co-authored a winning proposal for social housing resource integration platform, Taiwan
2017	Co-initiated a polylactic acid (PLA) cups recycling initiative, Netherlands (link)
2015	Contributed to drafting and lobbying of a nuclear waste management bill, Taiwan (pushed through first reading)

Skills

Programming languages and software
R, Python, HTML & CSS, Git, QGIS, Gurobi

Ecological and computational skills
High-performance computing, Topological data analysis, Linear programming, Phylogenetic comparative methods, Ancestral state reconstruction, Network analysis, Species distribution modeling, Spatial data analysis, Biodiversity measurement, Animal tracking data analysis

Languages Mandarin Chinese, English

Others WMA wilderness first responder, PADI advanced open water diver,
NCMW adult mental health first aid certification